

$$\text{and } \prod_{i=1}^{CN^2 d} \frac{1}{\beta} = \prod_{i=1}^{CN^2 d} \frac{1}{\beta}$$

$$\text{for } C = \frac{CN^2 d}{2}.$$

which is equal to

$$(e^{-\log \beta})^{CN^2 d} = e^{-CN^2 d \log \beta}$$

as desired $\square$

$$\frac{1}{Z} \leq e^{CN^2 d \log \beta}$$

$$\Rightarrow \frac{e^{-\beta S(u)}}{Z} d\sigma_n \leq$$

$$e^{CN^2 d \log \beta} \frac{e^{-\beta S(u)}}{Z} d\sigma_n$$

$$= Z \cdot e^{CN^2 d \log \beta} \frac{e^{-\beta S(u)}}{Z} d\sigma_n$$

$$\int_B \frac{\exp(-\beta S(u))}{Z} d\sigma_n(u)$$

$$\leq \int_B \frac{\exp(-CN^2 d \log \beta - t)}{Z} d\sigma_n(u)$$

because on the set $B = \{ \beta S(u) \geq CN^2 d \log \beta + t \}$,

we know that $-\beta S(u) \leq -CN^2 d \log \beta - t$

$\Rightarrow \exp(-\beta S(u)) \leq \exp(-CN^2 d \log \beta - t)$,
and the integral

$$\int_B \frac{(\cdot)}{Z} d\sigma_n(u) \text{ is monotone.}$$

Then, since the integral is positive,
increasing the numerator increases the value:

$$\leq \exp(-CN^2 d \log \beta - t) \int \frac{1}{Z} d\sigma_n(u)$$

$$= \exp(-CN^2 d \log \beta - t) \frac{1}{Z}$$

$$\leq \exp(-CN^2 d \log \beta - t) \exp(CN^2 d \log \beta) = e^{-t}$$

$$\mu_{YM}(BS(u) \geq Cn^d \log \beta)$$

$$= \int \frac{e^{-BS(u)}}{Z} d\omega_n(u)$$

u:

$$BS(u) \geq$$

$$Cn^d \log \beta$$

$$\leq \int \frac{e^{-Cn^d \log \beta}}{Z} d\omega_n(u)$$

u:

$$BS(u) \geq$$

$$Cn^d \log \beta$$

$$\leq \int_{U(B_n)} \frac{e^{-Cn^d \log \beta}}{Z} d\omega_n(u)$$

$$= \frac{e^{-Cn^d \log \beta}}{Z} \int_{U(B_n)} d\omega_n(u)$$

$$\leq \frac{e^{-Cn^d \log \beta}}{e^{Cn^d \log \beta}} = 1.$$

$$S(u) \geq Cn^d \frac{\log \beta}{\beta} + \frac{t}{\beta}$$

has no YM weight.

(actually less than e^{-t} YM weight)

Why? from the lower bound on the partition function (upper bound on the reciprocal),

we have an exponential bound on the normalizing constant for the YM measure, where the exponential ~~precisely cancels out~~ is of the form $e^{Cn^d \log \beta}$, matching the expression $BS(u) \geq Cn^d \log \beta$

$\beta S(u) \leq -Cn^d \log \beta - t$
has no QM weight.

the normalizing factor $\frac{1}{Z}$
for the QM measure is
upper bounded by
 $e^{Cn^d \log \beta}$.

Then

$$\frac{\exp(-\beta S(u))}{Z} \leq \frac{\exp(Cn^d \log \beta)}{\exp(-Cn^d \log \beta - t)}$$

QM density $= e^{-t}$

The QM density is upper bounded
by e^{-t} , when
 $S(u)$ exceeds $n^d \log \beta / \epsilon^{1/d}$.

How does the lower bound occur?

$$\begin{aligned} \text{Circ. } \frac{1}{Z} &\leq e^{Cn^d \log \beta} \\ \text{a.e. } Z &\geq e^{-Cn^d \log \beta} \\ &= \left(\frac{1}{\beta}\right)^{Cn^d} \end{aligned}$$

~~Wilson action~~

Contribution of each plaquette to
Wilson action is on a d g smaller
than $1/\beta$.

($\rightarrow$) this reduces to small ball
probabilities of Haar measure.

$$Z \geq \int_{B_1} \int_{B_1} \int_{B_1} d\omega d\omega d\omega$$

$$= \int_{\frac{B_1}{2\sqrt{d}}} \int_{\frac{B_1}{\sqrt{d}}} \int_{\frac{B_1}{1/\sqrt{d}}} d\omega d\omega d\omega$$

$$1 = \frac{1}{Z} \int_{\mathcal{U}(\beta n)} e^{-\beta S(u)} d\sigma(u)$$

$$Z = \int_{\mathcal{U}(\beta n)} e^{-\beta S(u)} d\sigma(u)$$

- Chinatown
- back to black
- as for me
- all we imagine as light
- elufak that was Stones
- goodfellas
- habemus papam (we have a pope)
- Harry Potter
- inception
- jim knopf
- jurassic...

- Leonardo in the morning light
- love is outlabeled
- the godfather
- the matrix

The Godfather

"went to the police like a good american"

I can become sad like them if I put my mind to it.

- fucking w) a dress on
- "bring a real man"
- a man who doesn't feel for w)
- his family can't be a real man"
- giving bad news immediately
- what does it mean to be a serious man?

Section 8.

$$\mu_{\text{FM}}(\text{set } A) \leq e^{-t}$$

for any t , where

$$A = \{u \in \mathcal{U}(B_n) : \beta S(u) \geq C n^d \log \beta + t\}$$

(i.e. the Yang Mills (abelian) of configurations with

Wilson action larger than $O(n^d \log \beta)$ is small).

$$\text{And as a corollary, } Z(B_n, g_0) \leq 2 \int_A e^{-\beta S(u)} d\mu_{B_n}(u)$$

$$\text{where } A = \{\beta S(u) \leq C n^d \log \beta\}$$

which is an upper bound on the partition function using the

energy bound under Yang Mills
lattice measure.

What is the lattice measure?

$$\mu_{\text{FM}}(\mathcal{U})$$

It is a measure on configuration space.

How does this define the graph measure?

Graph measure is a weight on vertices typically, not on configurations.

What does section 4 say?

It says:

$$Z \geq e^{-C n \log \beta}$$

This is based on assuming
~~these~~ average contribution
of a plane wave to the
Wilson action is small.

Then using axial gauge
fixing, this can be restated.

It also uses a small ball
probability estimate from section
6:

$$C_1 \delta^2 < O(\|I - U\| \delta) \\ < C_2 \delta^2.$$

for δ small enough based on
 N .

Dann mache ich es dir
zur Pflicht, diese Bestimmung
einzuhalten.

"Therefore I command you
to do this"

- Geheimnis $\approx$ secret?
- Pflicht $\approx$ duty

I make it your duty to

$$\begin{aligned} \bullet \quad SU(2) \ni A &\mapsto U(A) \\ &\quad \psi(R(A)^{-1} \cdot) \\ &\quad \forall \psi \in L^2(\mathbb{R}^3, \mathbb{C}^n) \end{aligned}$$

$$u: SU(2) \rightarrow U(\mathcal{H})$$

What is R ? it's the 2-1
map $SU(2) \rightarrow SO(3)$